Matrices

(P 1)

A discrete dynamial system is described by the following transition matrix, T,

$$T = \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.5 \end{bmatrix}$$

The state of the system is defined by the proportion of population with a particular characteristic.

a) Use the characteristic polynomial T to find its eigenvalues. [2]

b) Find the corresponding eigenvectors of T. [2]

c) Hence find the steady state matrix s of the system. [2]

a) The characterstic poly. of T is

$$\det(T - \lambda I) = \begin{vmatrix} 0.6-\lambda & 0.5 \\ 0.4 & 0.5-\lambda \end{vmatrix}$$

$$= \lambda^2 - 1.1\lambda + 0.1$$

$$\det(T-\lambda I) = 0$$

$$\lambda_1 = 0.1, \quad \lambda_2 = 1$$

b) Let X_1 and X_2 be the corresponding eigenvectors. For $\lambda_1 = 0.1$, we have.

$$(T - \lambda_1 I) X_1 = 0$$

$$\begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$0.5a + 0.5b = 0$$

$$a + b = 0$$

$$a = -b, \quad \text{So} \quad X_1 = \begin{bmatrix} -b \\ b \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$